

AVISH PRADHAN

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Education

Year	Degree / Exam	Institute	GPA / Marks
2023 – 2027	B.Tech in Computer Science	Graphic Era Deemed University, Dehradun	8.35
2022	Class XII (CBSE)	The Avenue Public School, Meerut	93.6%

Skills

- **Programming:** C++, Python, Java, SQL
- **Web Development:** Flask, REST APIs, PostgreSQL, SQLite, HTML, CSS, JavaScript, Bootstrap
- **Core Concepts:** Data Structures & Algorithms (150+ problems solved), Authentication (JWT, TOTP), Session Management
- **Tools:** Git, GitHub, Render

Projects

SecureAuth — Two-Factor Authentication Web App

Flask, PostgreSQL, bcrypt, JWT, TOTP, HTML/CSS

GitHub

- Engineered a Flask-based 2FA system using **TOTP** and **bcrypt**, securing authentication for 50+ simulated users.
- Implemented **JWT-based** access control reducing unauthorized route access to 0 during penetration testing.
- Optimized login response time to under 300ms through efficient session handling.
- Wrote modular authentication decorators reducing repeated backend code by **30%**.

Route Wise AI—AI-Powered Multi-Stop Routing and Traffic Navigation System

Python, Flask, JavaScript, Genetic Algorithms

GitHub

- Engineered an intelligent traffic navigation system using Python, Flask, and JavaScript, optimizing multi-stop routes with Genetic Algorithms and 2-opt heuristics to reduce average travel distance by **18%**.
- Integrated **3 external APIs** (MapQuest, TomTom, OpenWeatherMap) to process real-time traffic and weather data across **1,000+ route simulations**.
- Developed a Scikit-learn regression model improving delay prediction accuracy from **74% to 91%**.
- Optimized backend computation achieving route calculation times under **10 seconds** with responsive mobile-friendly UI support.

Real-Time Learning Scheduler

Flask, SQLite, Chart.js

GitHub

- Designed an adaptive study planner managing 50+ academic goals with dynamic task prioritization.
- Reduced scheduling conflicts by **25%** using Greedy and Knapsack optimization strategies.
- Processed **200+ simulated** scheduling scenarios to validate algorithm efficiency.
- Engineered a Flask-based backend with SQLite and real-time dashboard updates using Chart.js, improving schedule update latency to under 500ms.

Achievements & Certifications

- Solved **150+ DSA problems** on LeetCode covering Arrays, Binary Search, Recursion, Linked Lists, and Dynamic Programming.
- Strengthened backend engineering fundamentals through hands-on implementation of authentication, authorization, and REST API design in multiple projects.
- Completed AWS foundational modules including IAM, Shared Responsibility Model, and DevOps workflows on AWS.
- Actively participated in peer code reviews and debugging sessions resolving Flask routing, database integration, and backend flow issues.